Politecnico di Milano

School of Industrial and Information Engineering
M.Sc. in Aeronautical Engineering

ASSIGNMENT 1

Structural Dynamics and Aeroelasticity

Aeroelastic Analysis and Optimization of a Braced Wing Configuration

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1 Numerical Result Report

1.1 Divergence Dynamic Pressure Analysis(1.a)

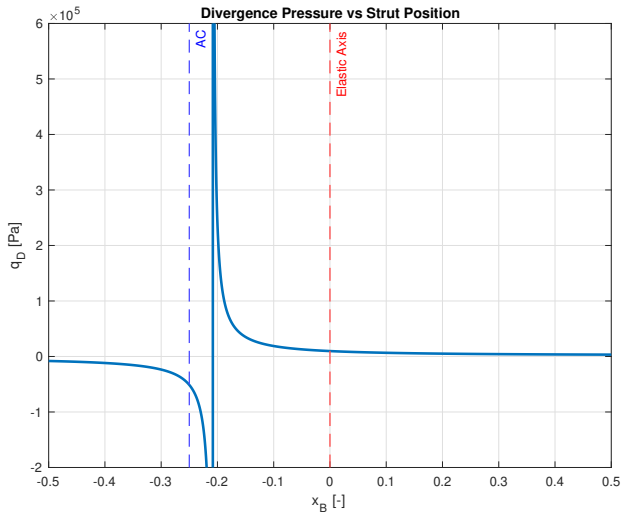


Figura 1: Evolution of q_D with respect to the chordwise coordinate of point B

The chordwise position of the strut attachment plays a key role in determining the torsional stability of the wing. It can be observed that as x_B moves forward toward the leading edge, the divergence dynamic pressure q_D becomes negative, indicating that the wing achieves complete immunity to static divergence, because q can assume only positive values.

1.2 Control Effectiveness and Strut Sensitivity Analysis (1.b)

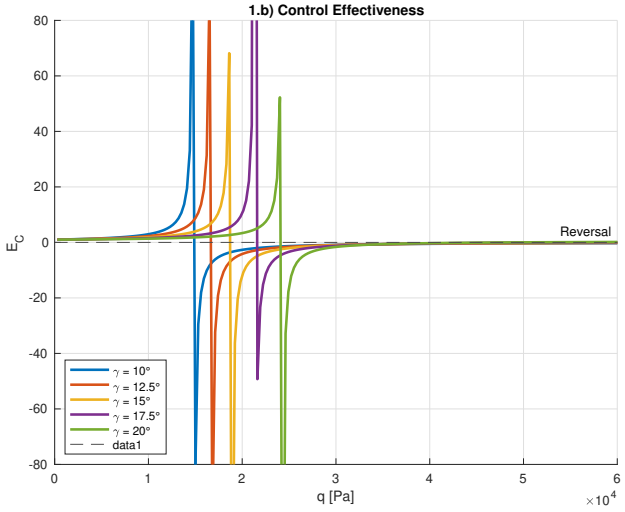


Figura 2: E_c evaluated over dynamic pressure q

The variation of control effectiveness E_c as a function of the dynamic pressure q is reported for different values of the strut inclination angle γ. It can be observed that, for all configurations, E_c initially remains positive and gradually increases with q, indicating a stable and effective control response. However, as the dynamic pressure approaches a critical value, each curve exhibits a

vertical asymptote corresponding to the aeroelastic instability of the system. Beyond this point, the control effectiveness becomes negative, meaning that the aeroelastic twist counteracts the control surface action. This behavior is associated with the onset of control reversal.

1.3 Deformed Shape Comparison(1.c)

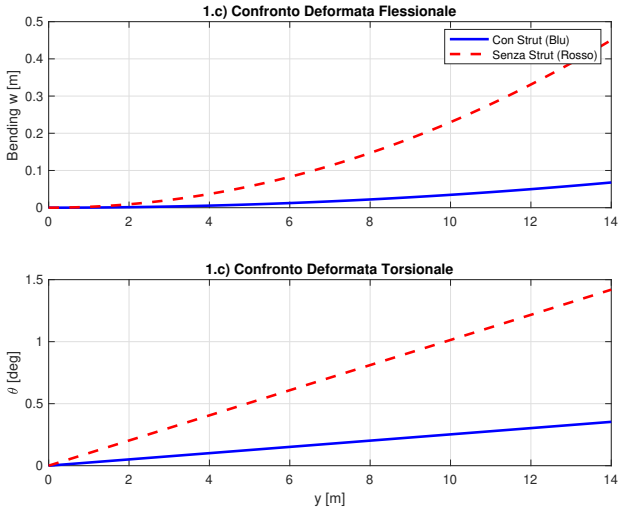


Figura 3: Bending and torsional shape comparison between the unbraced and braced configurations($q = 0.4 \cdot q_D$,$\beta_0 = 10deg$)

The blue lines demonstrate that, by adding the strut, both bending (top) and torsional deformation (bottom) are drastically reduced along the entire span to the unreinforced configuration (red dashed lines). In short, the strut effectively increases the structural stiffness, minimizing displacement and rotation.

1.4 Control Reversal Analysis(1.d)

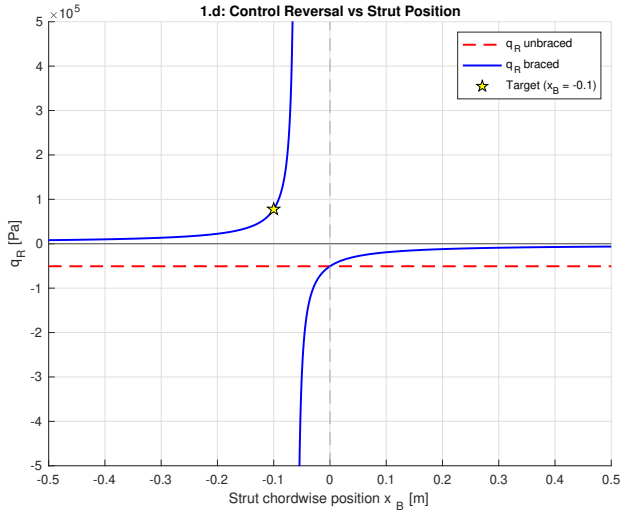


Figura 4: Dependency of the reversal dynamic pressure on the strut attachment point x_B

While the unbraced configuration remains constant, the braced structure shows a hyperbolic dependency on x_B, highlighting a critical asymptotic behavior where the chosen target position (yellow star) significantly enhances the reversal threshold compared to the baseline.

1.5 Numerical Convergence and Ritz-Galerkin Formulation(1.e)

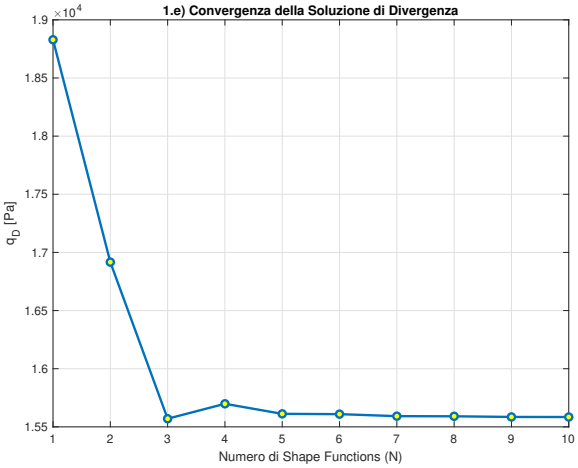


Figura 5: Convergence behaviour of the divergence dynamic pressure against the polynomial basis dimension N

This plot demonstrates the convergence of the divergence dynamic pressure (q_D) as a function of the number of shape functions. The system achieves a stable solution rapidly, indicating that using 3 to 5 shape functions is sufficient to reach accurate numerical convergence.

1.6 Design Optimization and Mass Minimization(1.f)

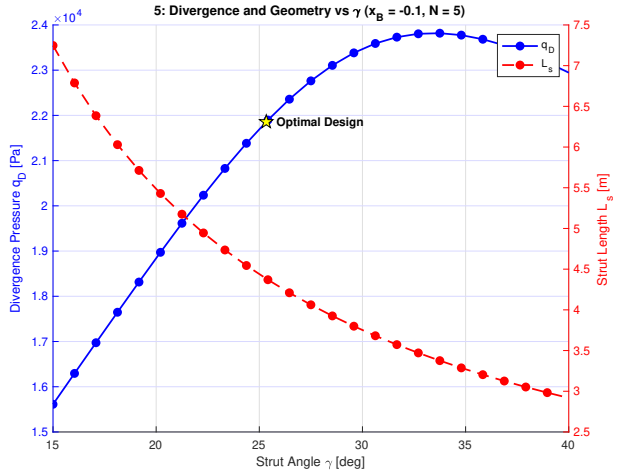


Figura 6: Impact of the strut angle γ on divergence dynamic pressure (blue curve) and required strut length L_{strut}. The star indicates the optimal configuration

This graph demonstrates that increasing the strut angle (γ) effectively improves the divergence pressure (q_D) while simultaneously reducing the required strut length (L_s). The highlighted optimal design point represents a 40% increase in q_D, illustrating a highly efficient structural trade-off.